# Abhirup Paul

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## Summary

Backend software engineer building well-architected, scalable enterprise-grade solutions across high-traffic, real-time systems — data analytics services, payment orchestration, and event-driven microservices. Hands-on experience in **Golang** (concurrent data pipelines, analytics services) alongside strong **Python** backend depth. Comfortable with software design patterns, Git, Docker/Kubernetes, SQL databases, and CI/CD (GitLab CI, GitHub Actions); cross-platform development on Linux and Windows. Co-inventor of a patented clinical platform (Neuroscan360) deployed in top-tier hospitals.

## Education

**GKV FET, Haridwar** — Uttarakhand, India
*B.Tech, Electronics and Computer Engineering — CGPA: 8.5 / 10.0* — *2022 – 2026*

- Research association — **SCAN Lab, IIT Bombay**, under Prof. Samir K. Maji (Dept. of Biosciences & Bioengineering); applied machine learning and deep learning to biomedical signal processing for neurological assessment.

## Experience

**R&D Engineer — SCAN Lab, IIT Bombay** — Mumbai, India
*Indian Institute of Technology Bombay* — *Jul 2025 – Present*

- Architected **Neuroscan360**, a patented full-stack clinical screening platform for neurological assessments, live in Lilavati Hospital and Kokilaben Dhirubhai Ambani Hospital; used daily by multiple medical teams.
- Built scalable Python server applications (FastAPI) and a **Golang** data-analytics service for batch processing of multimodal patient records; containerised all services with **Docker**, orchestrated on **Kubernetes (k8s)**, and shipped via CI/CD (GitLab CI, GitHub Actions) with cross-platform support across Linux and Windows.
- Built the cross-platform React Native mobile app (iOS & Android) from scratch — device sensor integration, real-time multimodal data capture, and clinical workflow UX; pending App Store & Play Store publication.
- Designed FastAPI microservices with **sub-50ms inference latency**, serving real-time ML models for neurological assessment over REST and WebSocket.
- Owned end-to-end PostgreSQL (SQL) schema for multimodal patient data; deployed on AWS Well-Architected Framework — improved system reliability by **64%** and cut infrastructure cost by **30%**.

**Freelance Backend Engineer** — Remote
*Independent* — *Dec 2024 – Feb 2025*

- Designed and shipped a rule-based **fraud signal engine** processing real-time financial transactions — full ownership of API design, data modelling, event processing, and infrastructure.
- Built a WebSocket-based real-time alerting system with ledger-aware backend ensuring transaction integrity and preventing duplicate/replayed events; secured APIs with JWT authentication.
- PostgreSQL schema + Redis caching for low-latency lookups; chose Redis Pub/Sub over a managed queue to cut latency and operational overhead. *Stack: Python, FastAPI, PostgreSQL, Redis, AWS*

## Projects

**Event Ticketing Management System** | *Python, FastAPI, PostgreSQL, Redis, WebSocket, Stripe, Docker*

- Built a full-stack event ticketing platform with **idempotent payment orchestration** (Stripe), QR-code ticket generation, and real-time seat inventory updates via WebSocket to prevent overselling under concurrent load.
- Designed a state-machine-backed booking lifecycle (draft → reserved → confirmed → cancelled) with Redis distributed locks ensuring race-condition-free seat reservation; **SQL (PostgreSQL)** schema with full audit trail.
- Engineered an async notification pipeline (email + in-app) triggered by booking events; Dockerised microservices with CI/CD via GitHub Actions.

**Real-Time Data Pipeline & Analytics** | *Golang, Redis, WebSocket, REST APIs*

- Built a high-throughput, scalable real-time pipeline in **Golang** processing **10,000+ updates/day** at sub-100ms latency — concurrently ingesting and normalising heterogeneous data formats with goroutines and channels, computing derived metrics, and detecting actionable signals across event streams.
- Applied clean software design patterns for testable code; used Redis for low-latency caching and pub/sub messaging, exposing results over a clean REST API.

**Speech Emotion Recognition (SER)** | *Python, TensorFlow/Keras, CNN-LSTM, librosa*

- Built a deep-learning speech emotion recognition system (B.Tech final-year project) achieving **90–95% accuracy** with a **CNN-LSTM** architecture trained on combined **RAVDESS, TESS, and CREMA-D** datasets for robust cross-corpus generalisation; engineered an audio feature pipeline (MFCCs, mel-spectrograms, chroma) with augmentation and hyperparameter tuning to handle class imbalance.

## Technical Skills

**Languages:** Golang (Go), Python, C++, Java, JavaScript, TypeScript, SQL

**Backend:** REST API design, Microservices, Event-Driven Architecture, Concurrency (goroutines/channels), WebSocket, Software Design Patterns, JWT

**Databases:** PostgreSQL (pgvector), MongoDB, Redis, Firebase, SQL Databases, Vector Databases

**DevOps & Platforms:** Docker, Kubernetes (k8s), GitLab CI, GitHub Actions, CI/CD, Git, Jira, Make, AWS (Well-Architected), Cloud Providers, Cross-platform (Linux & Windows), Distributed Systems

**AI / LLM:** LangChain, RAG Pipelines, LLM API Integration, Claude API, OpenAI API; AI-Native Dev Tools (Claude Code, Cursor, GitHub Copilot)

**Mobile & Frontend:** React Native (iOS & Android), React.js, Next.js, Redux

## Achievements

Co-inventor — **Neuroscan360** patent (IIT Bombay) | Runner-up — **MumbaiHacks 2024**